The School of Mathematics



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Predicting Used Tractor Auction Prices

Exploratory Analysis and Predictive Modelling

by

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Executive Summary

This project investigates the factors associated with used-tractor prices and develops a model for estimating realised sale prices from a client-supplied listings dataset. The original data contained listings from both online marketplaces and auction platforms. Marketplace records predominantly contained asking prices rather than confirmed transaction prices, so the modelling sample was restricted to auction records with a recorded sold price. After applying data-quality checks and removing repeated observations of the same tractor, the final modelling sample contained 22,857 unique tractors. Exploratory analysis showed strongly right-skewed prices, large differences in market composition across currencies, and clear relationships between price and characteristics such as manufacture year, usage and tractor model.

The exploratory analysis also determined which variables could be used reliably for modelling. The final models used features with sufficient coverage, including manufacturer, model series, manufacture year, usage and market information. Some potentially informative variables were too incomplete to include. Horsepower was available for only **12.9%** of tractors, despite showing a clear positive relationship with price, while transmission was recorded for only **16.0%** and its categories also showed noticeable price differences. Their exclusion therefore reflects limited data coverage rather than a lack of pricing information.

Several modelling approaches were compared. Linear and quadratic regression were first used as reference models, followed by a Decision Tree, XGBoost and CatBoost. Model selection was carried out using a separate validation sample, where CatBoost performed best. The final CatBoost model was then refitted using the combined training and validation data and evaluated once on the previously unused test sample. Quadratic regression, which was the stronger of the two regression models, was used as the main comparison.

On the test data, CatBoost achieved a **MedAPE (median absolute percentage error) of 32.0%**, compared with **40.2%** for quadratic regression. In practical terms, the typical CatBoost prediction was therefore about **32% away from the realised auction price**. This is an improvement of **8.2 percentage points**, equivalent to a **20.5% reduction in typical percentage error**, and **47.6%** of predictions were within 30% of the final sale price. Performance varied considerably across markets: MedAPE was lowest for EUR-denominated sales at **18.0%** and highest for GBP-denominated sales at **57.5%**. CatBoost performed better than quadratic regression in four of the five currencies examined. Monetary errors are reported separately by currency in the Results section, since errors measured in different currencies are not directly comparable.

The results show that the available listing information contains useful information about realised auction prices and that CatBoost improves meaningfully on the regression benchmark. However, accuracy depends strongly on how well a tractor is represented in the historical data: common model series had a MedAPE of **22.6%**, compared with **44.0%** for rare or previously unseen series. The model should therefore be used to support valuation and identify plausible price ranges rather than as an automatic pricing rule. Better coverage of horsepower and transmission, a reliable auction-date variable, and more historical observations for less common tractor models would be the most useful improvements to the current system.

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University of Edinburgh – Own Work Declaration

Name: Fakhruddin Hussain

I confirm that all this work is my own except where indicated.

Date: 14th August 2026

Generative AI Disclosure

Generative AI was used as a supporting tool during this project and while preparing the report.

- **Report preparation:** ChatGPT was used to help organise sectionsand improve grammar and wording.
- **Coding and analysis:** ChatGPT was used to assist with selecting an appropriate evaluation metric and brainstorming approaches for transforming prices.

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1 Introduction and project context

1.1 Motivation and research question

Valuing used tractors is difficult because machines that appear similar can differ in age, usage, model, power, drivetrain, transmission and installed equipment. For an auction or equipment business, the practical question is predictive: given the structured information recorded for a tractor, how close can an estimated price be to the realised auction price, and where is that estimate likely to be unreliable?

The project was initiated from a client-supplied dataset containing agricultural-equipment listings from both online marketplaces and auction platforms, together with a practical business question: how can the available listing information be used to understand and predict realised tractor sale prices? The initial work therefore focused on understanding the structure and quality of the raw data, distinguishing asking prices from realised sale prices, identifying repeated listings, auditing missing fields and examining the main patterns associated with price. This exploratory stage was necessary both to define an appropriate modelling sample and to identify the variables that could be used reliably for prediction.

The central research question is:

> *How accurately can realised used-tractor auction prices be predicted from the recorded machine characteristics, which data limitations most constrain that accuracy, and what additional information would most improve a practical valuation model?*

1.2 Objectives

1. Construct a reproducible sample of unique realised tractor sales and document all material cleaning assumptions.
2. Carry out a substantive exploratory data analysis (EDA) of price distributions, currencies, missingness, age, usage, manufacturer, model series and recorded equipment.
3. Assess the completeness, consistency and predictive usefulness of the available variables to determine which features can be used reliably for modelling.
4. Compare regression benchmarks with nonlinear tree-based models on fixed training and validation samples.
5. Evaluate the selected model on an untouched test sample and communicate prediction errors in both percentage terms and actual sale-price units for each currency.
6. Translate the statistical results into data-quality and modelling recommendations that are useful to a company user.

1.3 External literature used for context

Hedonic price models provide a useful framework for understanding how the observed price of a differentiated product is related to its bundle of characteristics (Rosen, 1974). Tractor-market studies have similarly found that age, usage, machine characteristics and selling environment can be related to realised prices (Diekmann et al., 2008; Feuz, 2023). A recent study using 6,821 US auction observations of 300–450 horsepower tractors also reports nonlinear usage-price relationships and the relevance of detailed machine and market information (Pearson et al., 2026). These findings provide useful context for interpreting the patterns observed in the present exploratory analysis and the choice of predictive features.

2 Data preparation and exploratory analysis

The EDA is a central part of the project rather than a short preprocessing step. It is used to establish which observations represent the target business outcome, reveal missingness and category sparsity, identify plausible price relationships, and determine which variables can be used responsibly in the predictive model.

2.1 Sample construction and duplicate handling

The client-supplied raw dataset[1] contained 354,607 listings and 46 fields from both classified marketplaces and auction platforms. Before constructing the modelling sample, it was necessary to determine which observations contained a realised transaction price rather than a listed asking price.

The distinction is captured primarily by the `price_type` field. All 299,586 classified observations were labelled `price_type = asking`, whereas all 55,021 observations from the two auction channels, `auction_rba` and `auction_cheffins`, were labelled `price_type = sold`. The separate `status` field could not be used as a reliable indicator of a realised sale: only 20 classified listings had `status = sold`, while auction observations were recorded as `status = removed` despite having a sold price. This indicates that `status` describes the state of the observed listing rather than consistently identifying whether the displayed price is a transaction price.

Since the response of interest is a realised tractor sale price, classified marketplace observations were not used for model fitting. Their displayed values are asking prices, and removal of a listing does not establish that the tractor sold at that amount. The modelling sample was therefore restricted to tractor observations from the two auction channels with `price_type = sold`, producing 40,326 candidate rows before the remaining validity checks and duplicate removal.

The `observed_date` field records when a listing was scraped, not the auction date. The same sale can therefore be observed on several scrape dates. Scrape identifiers and scrape-specific metadata were excluded from the duplicate key, whereas model series and the substantive tractor characteristics were retained. Nine rows failed basic validity checks because manufacture year exceeded the 2026 reference year. Duplicate removal then removed 17,460 repeated records, leaving 22,857 unique observations (table 1).

Table 1: Construction of the final modelling sample.

Stage	Rows	Removed at step
Raw listings	354,607	–
Sold auction tractors	40,326	314,281
After validity checks	40,317	9
After repeated-scrape removal	22,857	17,460

High prices were inspected but not removed solely because they were statistically unusual. In an auction dataset, an expensive machine can be a genuine observation rather than an error. This avoids using an automatic IQR rule that would systematically discard a real upper segment of the market.

2.2 Missingness and feature availability

Missing values were treated according to field meaning. Under supplied domain guidance, missing drive values were mapped to `4WD`. Usage hours were missing for 25.1% of unique sales but were retained because coverage remained sufficient; missing values are median-imputed using training data only, with a separate indicator recording whether hours were originally unavailable. Horsepower and transmission were missing for 87.1% and 84.0% of observations, respectively, and were therefore excluded from the main model while retained for EDA.

The equipment fields require a different interpretation. In the raw data, these variables contain `True` when a feature is recorded and are otherwise blank. A blank cannot reliably distinguish genuine absence from

[1] Client-supplied `market_listings_new.csv` dataset. The data are not publicly distributed as part of this report.

non-reporting. For modelling, recorded `True` values are encoded as 1 and blanks as 0, so these variables should be interpreted as indicators that a feature was recorded rather than definitive presence/absence measures.

Table 2: Treatment of important fields with incomplete or specially encoded values.

Field or field group	Treatment
Usage hours	Retain; training-data median imputation plus a missingness indicator.
Boolean equipment indicators	Encode recorded `True` as 1 and blank as 0; interpret 0 as "not recorded", not confirmed absence.
Horsepower	Recorded for 2,947 tractors (12.9%); exclude from the main model because coverage is too sparse, but retain for EDA.
Transmission	Recorded for 3,666 tractors (16.0%); exclude from the main model because coverage is too sparse, but retain for EDA.

2.3 Price distribution and currency composition

Auction price is strongly right-skewed (figure 1). Taking logs compresses the extreme upper tail and makes relative differences more stable. The model therefore uses log price internally, but final prediction errors are transformed back to actual price units to make the results easier to interpret.

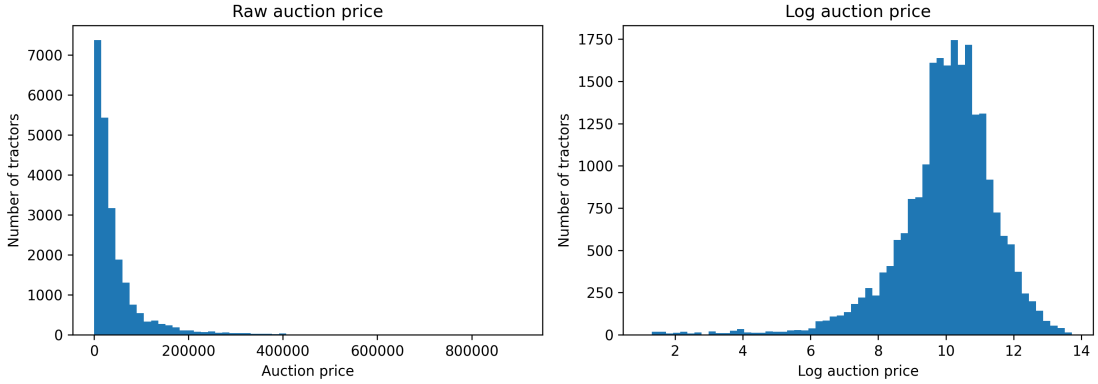


Figure 1: Raw and log auction-price distributions in the final unique-sale sample. The log transformation reduces the dominance of the upper tail without deleting genuine high-price sales.

The dataset contains five currencies with very different price levels and sample sizes (table 3). CAD and USD dominate the data, while AUD has only 442 unique observations. This is important both statistically and operationally: the model sees much less AUD information, and a numerical error of 10,000 has a different meaning in a low-price GBP segment than in a high-price CAD or AUD segment.

Table 3: Price distribution by currency in the full unique-sale sample. Values are in the native currency shown in the first column.

Currency	n	Median	25th pct.	75th pct.
CAD	8,461	39,000	16,000	82,500
USD	8,266	23,500	13,000	42,000
EUR	3,306	26,000	13,500	47,000
GBP	2,382	6,351	1,800	17,000
AUD	442	49,000	23,500	96,750

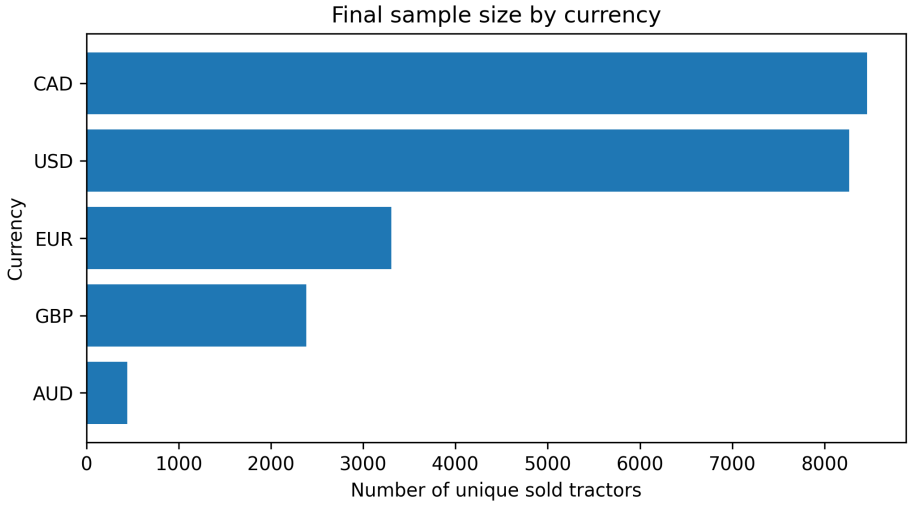


Figure 2: Number of unique sold tractors by currency. Currency-specific error estimates for smaller groups, especially AUD, should be interpreted with correspondingly greater sampling uncertainty.

For cross-currency EDA only, the report uses a currency-centred relative log price,

$$r_i = \log(P_i) - \text{median}\{\log(P_j) : c_j = c_i\}, \tag{1}$$

where P_i is sale price and c_i is currency. A value above zero indicates a tractor priced above the median tractor in the same currency. This measure is used only to compare relative price patterns across currencies; it is not a currency conversion and is not used as the final model metric.

2.4 Age, usage and price

Manufacture year shows a strong positive association with relative price: later-model tractors are generally more expensive. Usage hours show the opposite broad pattern, but with considerable dispersion, which is plausible because hours interact with age, model, equipment and condition. The scatter also suggests that one straight-line effect is unlikely to describe the full usage-price relationship.

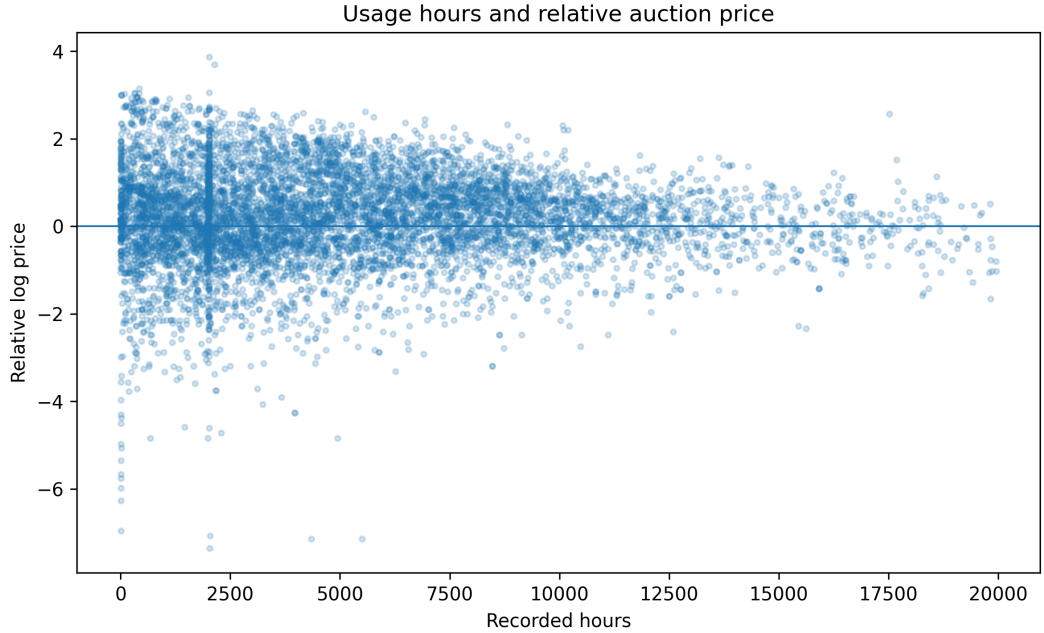


Figure 3: Recorded usage hours against currency-centred relative log price. The broad downward tendency is accompanied by substantial heterogeneity.

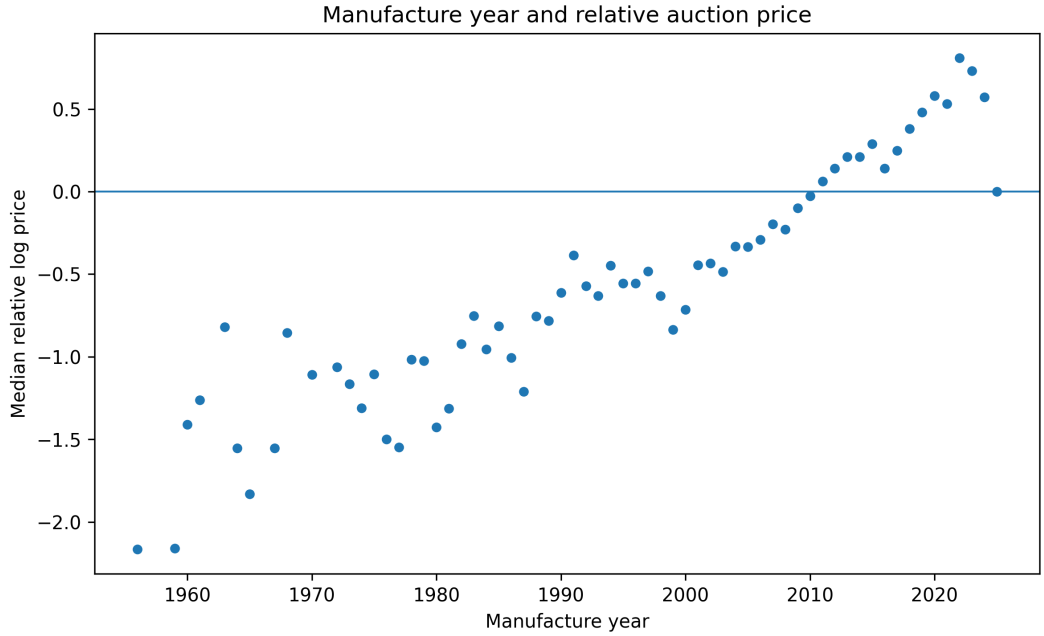


Figure 4: Manufacture year against median currency-centred relative log price for years with at least 20 observations. Newer machines generally occupy higher price positions within currency.

2.5 Horsepower and price

Horsepower is available for only 2,947 tractors (12.9%), but the observed subset shows a clear positive relationship with price (figure 5). The Spearman correlation between horsepower and currency-centred relative log price is 0.580, indicating that higher-horsepower tractors tend to sell at higher prices relative to other tractors in the same currency. However, horsepower is missing for 87.1% of the final sample. Including it in the main model would therefore require extensive imputation and assumptions about the missing values that are difficult to justify. Horsepower was consequently excluded from the final model despite its clear association with price.

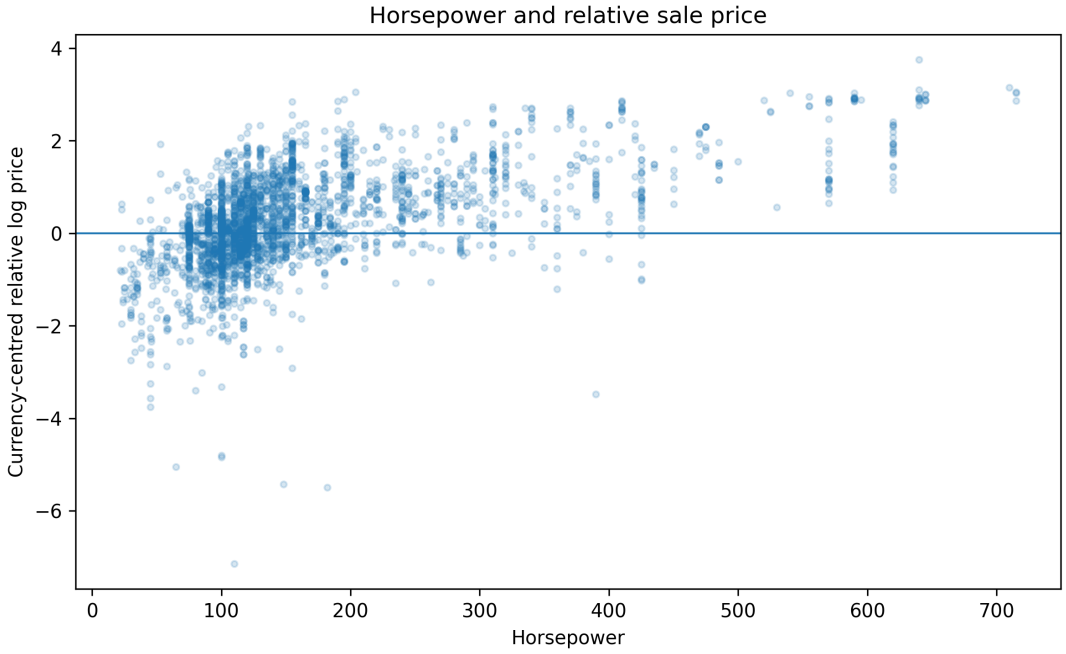


Figure 5: Horsepower against currency-centred relative log price for tractors with recorded horsepower. The plot shows a clear positive association between horsepower and price, although horsepower is available for only 12.9% of the final sample.

2.6 Transmission and price

Transmission is recorded for only 3,666 tractors (16.0%). Among the tractors where it is available, sale prices vary noticeably across transmission types (figure 6). Some transmission types tend to occur on higher-priced tractors, although these differences may also reflect manufacturer, model series, age and overall tractor specification. Since transmission is missing for 84.0% of the final sample and the recorded categories are not evenly represented across manufacturers and models, it was excluded from the final model. Nevertheless, the observed price differences suggest that more complete and consistently recorded transmission information could be useful for future modelling.

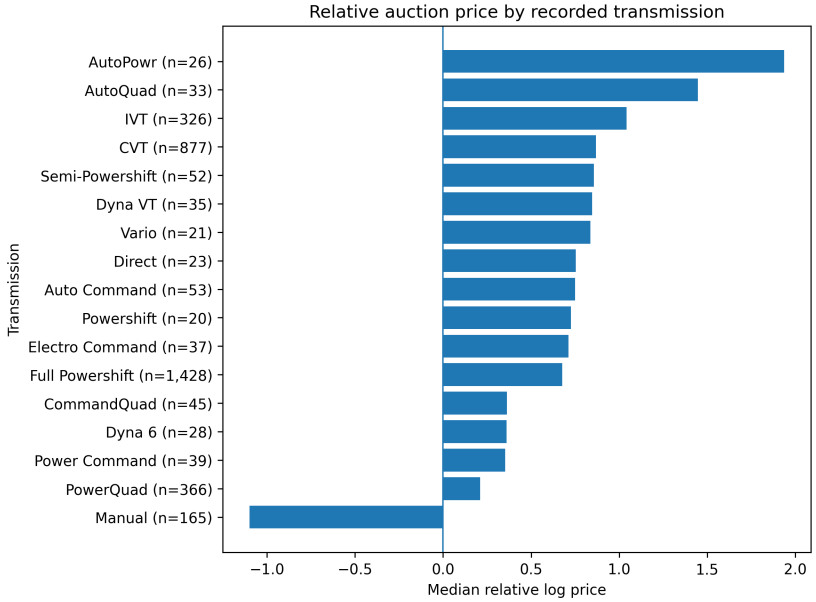


Figure 6: Median currency-centred relative log price by transmission type, shown for categories with at least 20 observations. Prices vary across transmission categories, although some of these differences may also reflect other tractor characteristics.

2.7 Manufacturer, model series and equipment

The final sample is dominated by a small number of manufacturers. John Deere accounts for 13,428 of the 22,857 tractors, followed by New Holland (3,549), Case IH (1,696) and Massey Ferguson (1,342) (figure 7). This means that the model has considerably more information available for some manufacturers than for others, which may affect how accurately prices can be predicted across different parts of the market.

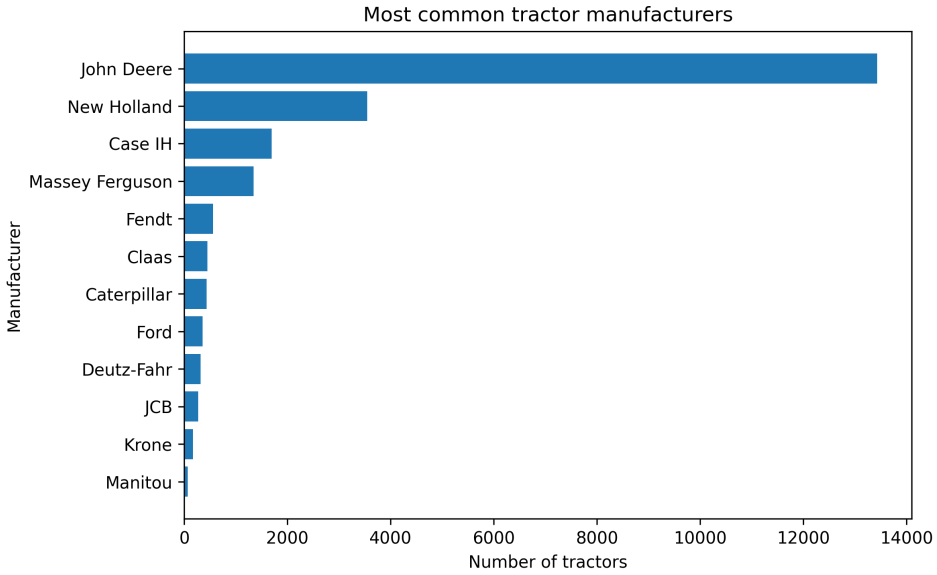


Figure 7: Most common manufacturers in the final sample. A large proportion of the available sales comes from a small number of manufacturers.

Model series are much more fragmented. The dataset contains 7,889 distinct model series, and many are represented by only a small number of sales (figure 8). In particular, 5,081 model series appear only once in the final sample. This is important for prediction because the model has limited historical information from which to learn the typical price of a rare or previously unseen series.

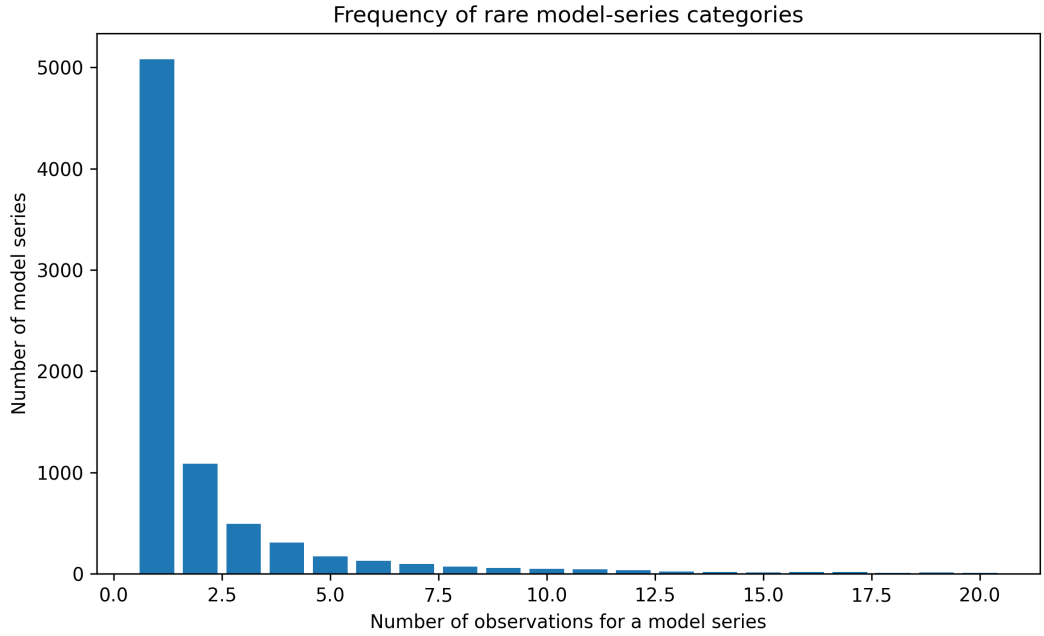


Figure 8: Number of observations available for individual model series. Many series are represented by very few historical sales, making their prices more difficult to estimate reliably.

The equipment indicators provide additional information about tractor specification. Features such as air conditioning, ISOBUS, cab suspension and rear-view cameras are more frequently recorded on tractors with higher prices relative to others in the same currency. These differences should not be interpreted as the direct effect of an individual feature, since equipment is also related to manufacture year, manufacturer and model series. However, the observed patterns suggest that the equipment indicators contain useful information for distinguishing tractors with different levels of specification and were therefore retained as predictors in the main model.

2.8 EDA implications for modelling

The EDA informed the final preprocessing and feature choices. Prices were modelled on a log scale and currency was handled explicitly because price levels differ across markets. Missing hours were retained using training-data median imputation and a missingness indicator. Equipment variables were treated as indicators that a feature was recorded, while rare model-series levels required grouping. Horsepower and transmission were excluded from the main model because their low coverage made extensive imputation difficult to justify despite their observed relationships with price.

3 Predictive modelling methodology

3.1 Train, validation and test design

The final sample of 22,857 tractors was split once into 70% training, 15% validation and 15% test sets using a fixed random seed. The split was stratified by currency and within-currency log-price quintile so that markets and broad price ranges were represented in all three subsets. A random rather than chronological split was used because the dataset does not contain a reliable transaction-date variable. The training set was used to fit the models and estimate preprocessing quantities, the validation set to compare models and hyperparameters, and the test set was kept untouched until final evaluation. The fixed seed makes the split reproducible.

Prices were modelled on the log scale to reduce the influence of the strong right-skew identified in the EDA. Because the dataset contains five currencies with different monetary units, log prices were then centred and scaled separately within currency using training-sample statistics. Converting all prices using exchange rates was avoided because reliable transaction dates are unavailable, so appropriate historical exchange rates cannot be assigned consistently.

For currency c, let m_c denote the median training log price and let

$$q_c = Q_{0.75,c} - Q_{0.25,c} \tag{2}$$

denote its interquartile range (IQR). The internal target is

$$z_i = \frac{\log(P_i) - m_{c_i}}{q_{c_i}}, \tag{3}$$

where P_i is the observed sale price and c_i its currency.

The median and IQR were used instead of the mean and standard deviation because they are less sensitive to extreme observations. Although logging reduces the original skew, unusually high and low prices remain, and the standard deviation can be strongly influenced by them. The IQR instead measures the spread of the middle 50% of log prices. Both m_c and q_c are estimated from the training data only so that the validation and test samples do not influence preprocessing.

The transformation standardises prices relative to the typical level and spread within each currency; it does not convert one currency into another. Predictions are returned to their original scale using

$$\widehat{P}_i = \exp\left(m_{c_i} + q_{c_i}\widehat{z}_i\right), \tag{4}$$

so monetary errors are reported in the observation's native currency, while percentage measures are used for cross-currency comparisons.

3.2 Additive linear regression

The first reference model is an additive linear regression. The response variable is the scaled log price z_i defined in equation (3). Let u_i denote the preprocessed manufacture-year value for tractor i and h_i its preprocessed usage-hours value. The categorical variables, including manufacturer, model series, country, currency and drive

type, are represented by dummy variables D_{ij}. The variables B_{ik} represent the equipment indicators and the indicator showing whether usage hours were originally missing. The model is

$$z_i = \beta_0 + \beta_1 u_i + \beta_2 h_i + \sum_{j=1}^{J} \gamma_j D_{ij} + \sum_{k=1}^{K} \delta_k B_{ik} + \varepsilon_i, \tag{5}$$

where β_0 is the intercept and ε_i represents the part of the scaled log price that is not explained by the included predictors. The coefficients β_1 and β_2 describe the relationship between the two continuous predictors and the response, while γ_j and δ_k measure differences associated with the categorical and indicator variables, respectively.

For a categorical predictor, one category is omitted and acts as the reference group. A coefficient γ_j therefore measures the expected difference in z_i between category j and the corresponding reference category, holding the other predictors fixed. Similarly, δ_k represents the difference associated with an indicator taking the value one rather than zero. Since the response is expressed on the scaled log-price scale, these coefficients should not be interpreted directly as changes in pounds, euros or dollars.

The coefficients are estimated by ordinary least squares (OLS). If $\widehat{z}_i$ denotes the fitted value for tractor i, OLS chooses the model coefficients to minimise the sum of squared residuals,

$$\min_{\boldsymbol{\theta}} \sum_{i=1}^{n} (z_i - \widehat{z}_i)^2 , \tag{6}$$

where $\boldsymbol{\theta}$ denotes the full set of regression coefficients. Squaring the residuals gives greater weight to larger errors and produces a single set of coefficients that gives the best least-squares fit to the training data.

This model provides a useful starting point because its structure is simple and each predictor enters the model in a clearly defined way. However, the continuous effects are assumed to be linear and the predictors contribute additively unless an interaction is included explicitly. For example, the model assumes that the relationship between usage hours and the response has the same form across manufacture years. If the effect of usage changes with tractor age, or if the relationship between year and price is curved rather than straight, the additive model cannot capture those patterns on its own.

3.3 Quadratic regression

The EDA suggested that the relationships between price, manufacture year and usage hours may not be adequately represented by straight lines. A quadratic regression was therefore fitted as a more flexible extension of the additive model. Using the same definitions of u_i, h_i, D_{ij} and B_{ik} as above, the model is

$$\begin{aligned} z_i = \beta_0 &+ \beta_1 u_i + \beta_2 h_i + \beta_3 u_i^2 + \beta_4 u_i h_i + \beta_5 h_i^2 \\ &+ \sum_{j=1}^{J} \gamma_j D_{ij} + \sum_{k=1}^{K} \delta_k B_{ik} + \varepsilon_i. \end{aligned} \tag{7}$$

The first two continuous terms, $\beta_1 u_i$ and $\beta_2 h_i$, have the same role as in the additive regression. The squared terms u_i^2 and h_i^2 allow the relationship with the response to bend rather than being restricted to a straight line. For example, a non-zero coefficient β_5 allows the change in predicted price associated with additional usage hours to differ between low- and high-usage tractors.

The interaction term $u_i h_i$ allows manufacture year and usage hours to influence each other. In the additive model, the effect of usage hours is assumed to be the same regardless of manufacture year. With the interaction included, the relationship between hours and the response can change depending on the age of the tractor. Similarly, the effect of manufacture year can vary according to the level of recorded usage.

This can be seen more directly from the partial effects of the two continuous predictors. Holding the categorical and indicator variables fixed, the change in the fitted response with respect to manufacture year is

$$\frac{\partial \widehat{z}_i}{\partial u_i} = \beta_1 + 2\beta_3 u_i + \beta_4 h_i, \tag{8}$$

while the corresponding change with respect to usage hours is

$$\frac{\partial \widehat{z}_i}{\partial h_i} = \beta_2 + \beta_4 u_i + 2\beta_5 h_i. \tag{9}$$

These expressions show why the quadratic model is more flexible than the additive regression. The relationship between year and price can change across different manufacture years and usage levels, and the relationship between hours and price can likewise change with both variables.

Despite the additional terms, the model is still a linear regression in the unknown coefficients $\beta_0, \ldots, \beta_5$, γ_j and δ_k. The term "quadratic" refers to the way the predictors enter the equation, not to the method used to estimate the coefficients. The parameters can therefore still be estimated using ordinary least squares.

The quadratic specification is intentionally limited to manufacture year, usage hours and their interaction. Creating squared terms and interactions for every manufacturer, model series and equipment variable would produce a much larger and harder-to-interpret model, particularly given the large number of rare model-series categories. The quadratic regression therefore provides a useful middle ground between the simple additive regression and the more flexible tree-based models considered next. Similar hedonic pricing studies have used squared continuous terms to represent nonlinear relationships (Feuz, 2023); here, however, the choice is motivated primarily by the patterns observed in the EDA of this dataset.

3.4 Tree-based models

The regression models impose a fixed mathematical form on the relationship between the predictors and price. However, the EDA suggests that some relationships may be nonlinear or depend on combinations of variables. Tree-based models are useful in this setting because they can learn such patterns without specifying every interaction in advance.

A regression tree repeatedly splits the data into smaller groups using predictor values. For example, a split may first separate newer from older tractors and then divide those groups further by usage, manufacturer or another characteristic. The final prediction is based on the group in which a tractor ends up. This allows trees to capture threshold effects and interactions naturally (Breiman et al., 1984).

Because a single tree can be unstable and prone to overfitting, two boosted tree methods were also considered. Gradient boosting combines many relatively small trees, with each new tree improving the errors left by the previous ones (Friedman, 2001). XGBoost is a regularised and computationally efficient implementation of this approach (Chen and Guestrin, 2016), while CatBoost is designed to handle categorical variables effectively (Prokhorenkova et al., 2018). This is particularly useful here because manufacturer and model series are important predictors and model series contains a large number of categories.

The tree-based models were not assumed to perform better simply because they are more flexible. Their usefulness was judged by whether they produced lower prediction error on the validation sample.

3.5 Evaluation in actual price units

Final performance is evaluated after transforming predictions back to the original sale-price scale. Within each currency, mean absolute error (MAE) is

$$\mathrm{MAE}_c = \frac{1}{n_c} \sum_{i:c_i=c} |P_i - \widehat{P}_i|, \tag{10}$$

where P_i is the realised price and $\widehat{P}_i$ is the predicted price. MAE is useful because it expresses the average prediction error directly in the relevant currency. However, it cannot be compared directly across currencies or across markets with very different price levels.

The main cross-currency measure is therefore median absolute percentage error (MedAPE),

$$\text{MedAPE} = \text{median}_i \left(\frac{|P_i - \widehat{P}_i|}{P_i} \right) \times 100. \tag{11}$$

MedAPE expresses each error relative to the realised price and is therefore comparable across currencies. It is preferred to mean absolute percentage error (MAPE) because very low-priced observations can produce extremely large percentage errors and disproportionately affect the mean. Taking the median makes the measure less sensitive to these extreme cases and gives a clearer indication of the error for a typical tractor. The proportions of predictions within 20% and 30% of realised price are also reported for easier interpretation.

Weighted absolute percentage error (WAPE) is reported only within individual currencies,

$$\text{WAPE}_c = \frac{\sum_{i:c_i=c} |P_i - \widehat{P}_i|}{\sum_{i:c_i=c} P_i} \times 100. \tag{12}$$

It is not pooled across currencies because this would require adding prices measured in different monetary units.

Uncertainty in the final CatBoost–quadratic regression comparison was quantified using 10,000 paired bootstrap resamples of the test observations, with the same sampled tractors used for both models in each replicate.

For the final test comparison, CatBoost is evaluated against quadratic regression. Quadratic regression performed better than the additive regression on the validation sample and therefore provides the stronger reference model. The currency-median predictor is retained only as a simple validation check.

4 Results

4.1 Regression diagnostics

On the validation set, back-transformed MedAPE was 38.1% for the additive regression and 38.0% for the quadratic regression, so the residual diagnostics use the slightly better quadratic specification. The residuals themselves are calculated on the scaled target used for model fitting. Residuals are broadly centred around zero, but their spread varies across fitted values, usage hours and manufacture years, with several large errors remaining (figures 9 to 11). This suggests that the quadratic regression captures part of the price structure but not all of the variation, motivating comparison with more flexible tree-based models.

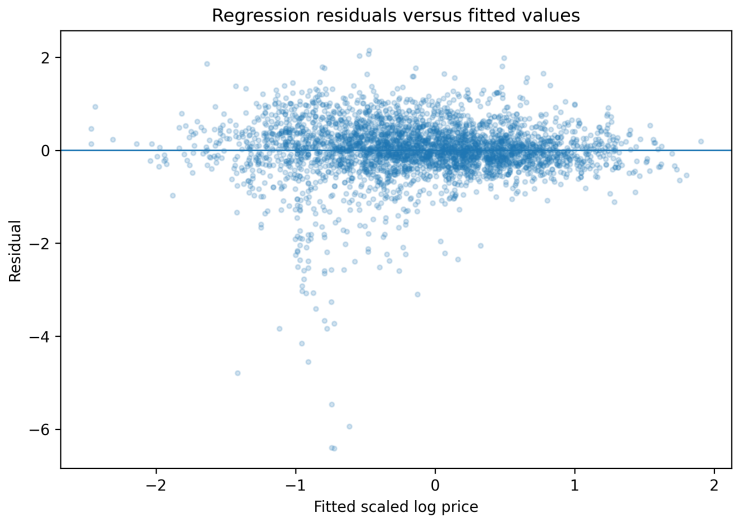


Figure 9: Validation residuals against fitted values for the quadratic regression. Changing spread across the fitted range indicates remaining heterogeneity.

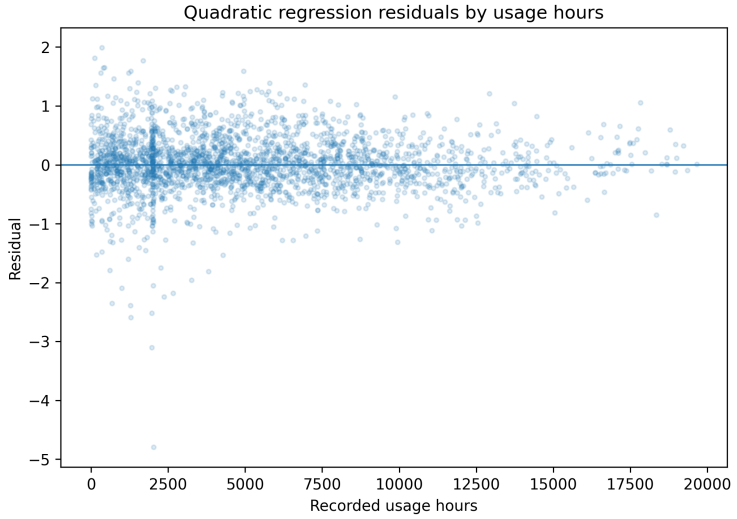


Figure 10: Validation residuals against recorded usage hours for the quadratic regression. Residuals are broadly centred around zero, although the spread is greater among tractors with lower recorded usage.

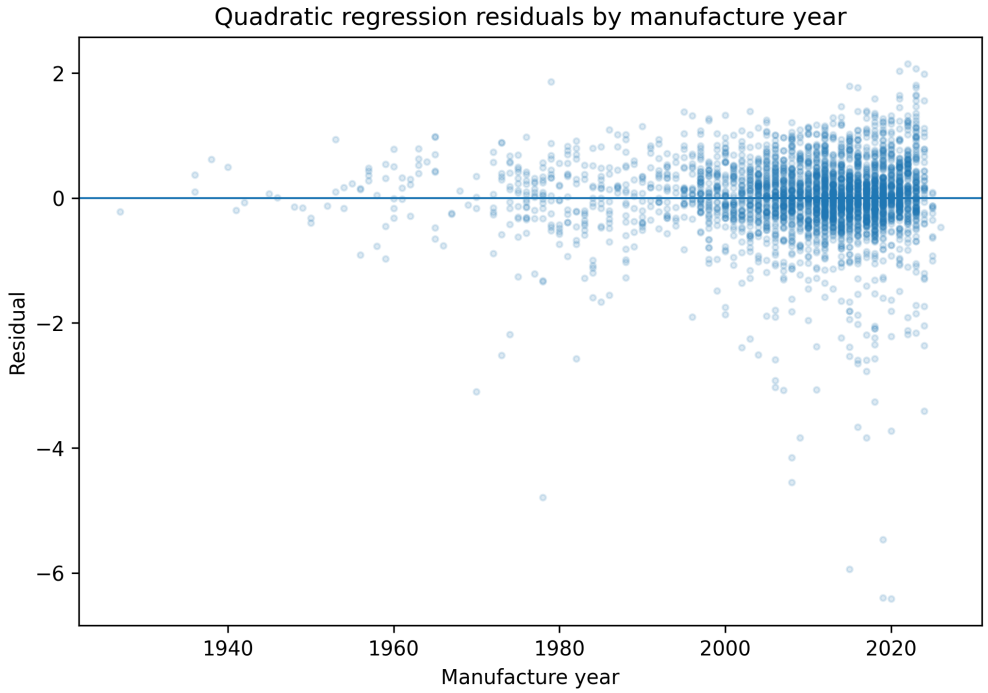


Figure 11: Validation residuals against manufacture year for the quadratic regression. Residuals remain broadly centred around zero, but their spread varies across manufacture years and several large errors remain.

4.2 Validation comparison and model selection

The models were compared on the validation sample before the test data were examined. Scaled MAE, calculated on the transformed target defined in equation (3), was used as the model-selection criterion because this is the scale on which the models were fitted. MedAPE and the proportion of predictions within 30% of the realised price were calculated after transforming the predictions back to their original price scale and are included to make the differences between models easier to interpret.

The currency-median predictor is included only as a simple reference to confirm that the fitted models provide useful predictive information beyond assigning a typical price within each currency.

Table 4: Validation performance used for model comparison. Scaled MAE is calculated on the transformed target, while MedAPE and "within 30%" are calculated after predictions are transformed back to the original price scale. Lower scaled MAE and MedAPE are better.

Model	Scaled MAE	MedAPE (%)	Within 30% (%)
Currency median reference	0.669	62.1	23.5
Additive linear regression	0.412	38.1	41.2
Quadratic linear regression	0.408	38.0	41.4
Decision Tree	0.394	35.7	43.7
XGBoost	0.371	34.9	44.3
CatBoost	**0.352**	**30.5**	**49.5**

CatBoost achieved the lowest scaled MAE on the validation sample and also produced the lowest MedAPE and the highest proportion of predictions within 30% of the realised price. The tree-based models generally performed better than the regression models, suggesting that the additional nonlinear relationships and interactions captured by these models improve prediction accuracy.

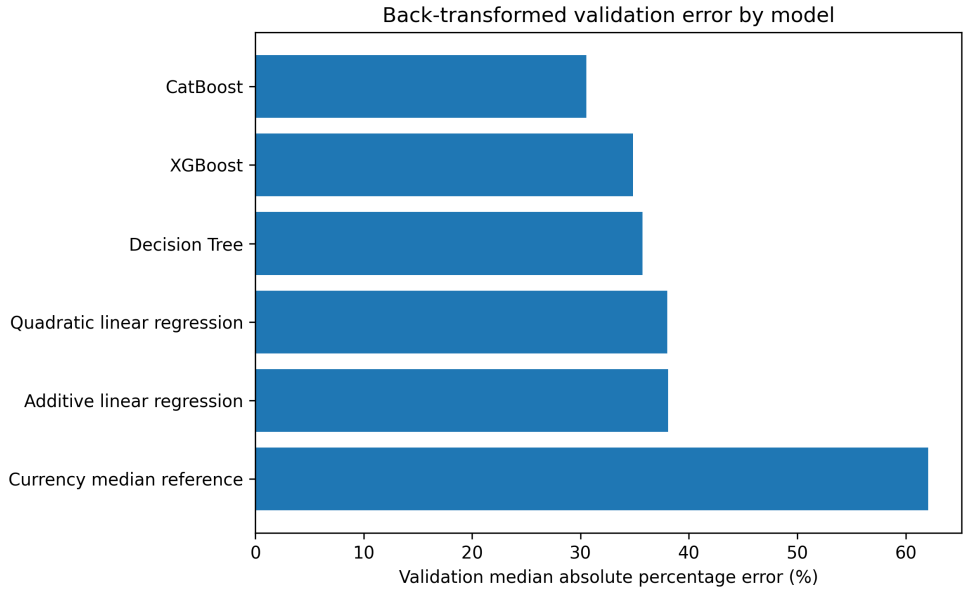


Figure 12: Validation model comparison using MedAPE calculated after back-transforming predictions to the original price scale. CatBoost achieved the lowest validation MedAPE.

CatBoost was selected using scaled MAE before the test set was examined. The selected model was then refitted on the combined training and validation data using the chosen hyperparameters before being evaluated once on the held-out test sample.

4.3 Held-out test performance in actual price units

The final test results are shown in table 5. After model selection, both CatBoost and the quadratic regression were refitted using the combined training and validation data before being evaluated on the untouched test sample. The quadratic regression is used as the main comparison because it was the stronger of the two regression models on the validation data.

Native-currency MAE is reported to show the monetary size of the prediction errors within each market, while MedAPE is used to compare accuracy across currencies. The "relative reduction" column shows the percentage reduction in MedAPE achieved by CatBoost compared with quadratic regression; a negative value therefore indicates that CatBoost performed worse.

Table 5: Final CatBoost test performance by currency. MAE and median price are shown in the native currency, while MedAPE is used to compare relative prediction accuracy across markets.

Currency	n	Median price	CatBoost MAE	CatBoost MedAPE (%)	Quadratic MedAPE (%)	Relative reduction (%)	Within 30% (%)
AUD	66	45,000	31,620	35.5	34.4	-3.3	43.9
CAD	1,270	38,000	29,225	40.0	50.4	20.6	39.2
EUR	496	26,000	7,594	18.0	24.9	27.6	68.1
GBP	357	6,400	7,373	57.5	71.5	19.5	24.1
USD	1,240	23,000	12,793	26.1	30.4	14.2	54.9

Across all 3,429 test observations, CatBoost achieved a MedAPE of **32.0%**, compared with **40.2%** for quadratic regression. This is an improvement of **8.2 percentage points** (95% paired-bootstrap CI: **6.6 to 10.1**), equivalent to a **20.5% reduction in typical percentage error**. In addition, **34.2%** of CatBoost predictions were within 20% of the realised price and **47.6%** were within 30%. These overall percentage-based measures can be calculated across currencies because each tractor's error is first expressed relative to its own realised price.

The currency-specific results also show why MAE should not be used on its own to compare markets. EUR and GBP have similar numerical MAEs, approximately 7,600 and 7,400 in their respective currencies, but their MedAPEs are very different at 18.0% and 57.5%. The median GBP sale price is much lower, so a similar-sized

monetary error represents a much larger proportion of the tractor's value.

CatBoost performed better than quadratic regression in CAD, EUR, GBP and USD. AUD was the only exception, where quadratic regression achieved a slightly lower MedAPE of 34.4% compared with 35.5% for CatBoost. However, the AUD test group contains only 66 observations, so this result should be interpreted with greater caution than the results for the larger currency groups.

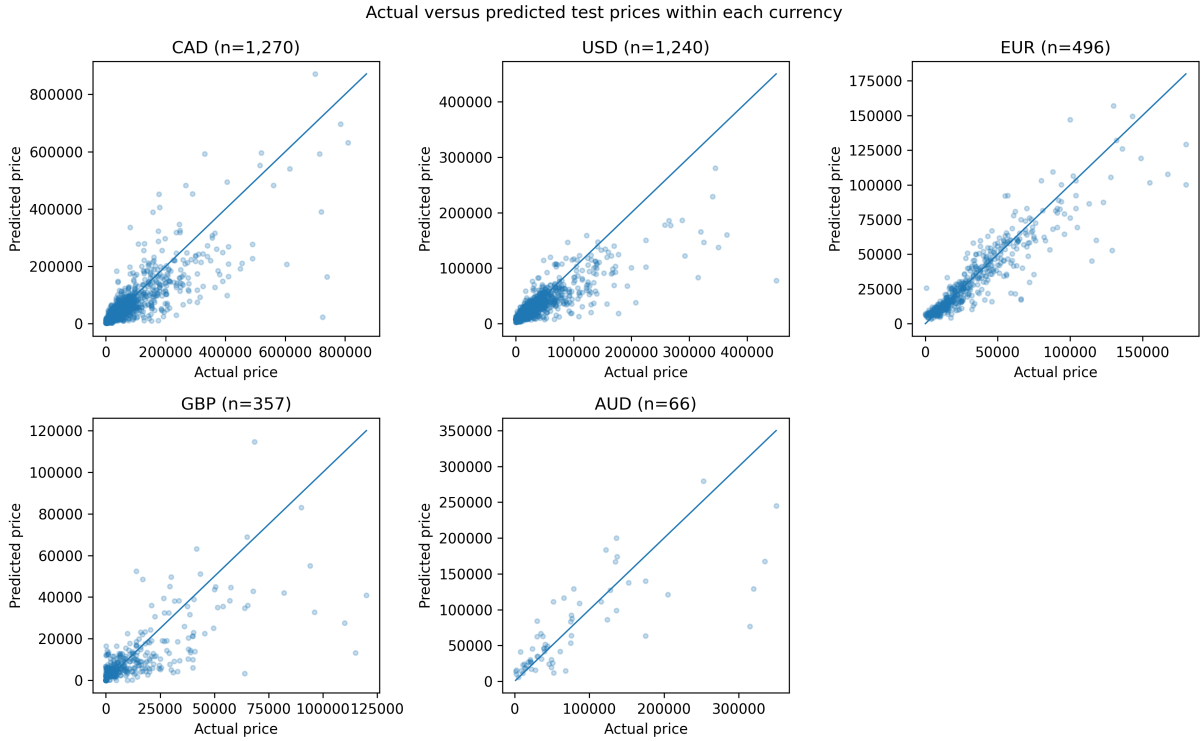


Figure 13: Actual and predicted test prices shown separately for each currency in native price units. The 45-degree line represents perfect prediction. Separate panels are used because the numerical price scales differ substantially between currencies.

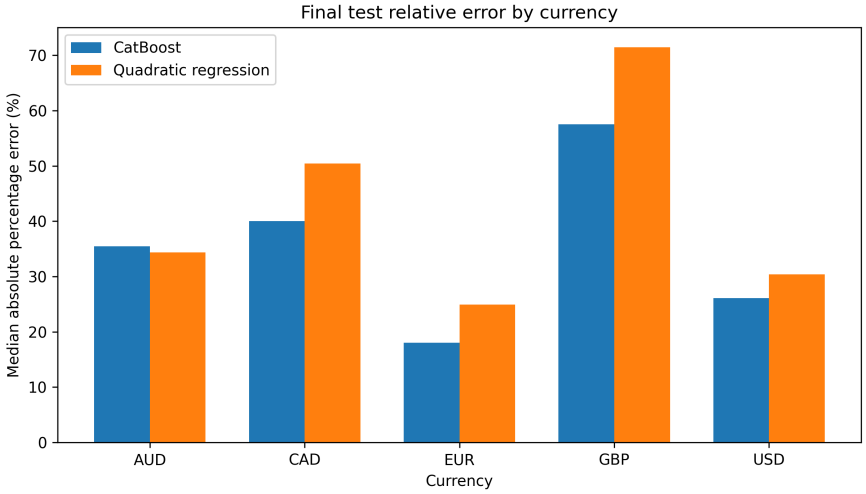


Figure 14: Test MedAPE by currency for CatBoost and quadratic regression. Because MedAPE expresses errors as percentages of the realised price, performance can be compared across currencies. CatBoost performs better in four of the five currency groups, with AUD as the exception.

4.4 Performance for common and rare model series

Model-series support is one of the clearest sources of performance variation. Tractors from model series sufficiently represented in the training data have MedAPE 22.6%, whereas rare or previously unseen series have MedAPE 44.0% (table 6). The share within 30% falls from 60.4% to 36.3% for rare or unseen series.

Table 6: Final test performance by model-series support. Per-observation percentage metrics are used because each subgroup contains multiple currencies.

Model-series support	n	MedAPE (%)	Within 20% (%)	Within 30% (%)
Common in training data	1,605	22.6	44.4	60.4
Rare or unseen	1,824	44.0	25.3	36.3

This is both a modelling and data-coverage result. When a model series has very few historical examples, the model has limited information from which to learn a reliable series-specific price pattern. For practical use, model-series frequency should therefore be retained as a reliability indicator alongside each prediction.

4.5 Predictive importance of recorded characteristics

Permutation importance measures how much the fitted model's predictive error increases when one feature is shuffled while the remaining test data are held fixed. It is a measure of model reliance rather than a causal effect (Fisher et al., 2019). Manufacture year produced the largest increase in internal MAE when permuted (0.123), followed by model series (0.097) and air conditioning (0.086). Manufacturer and currency were next, and the missing-hours indicator ranked slightly above the recorded hours value itself.

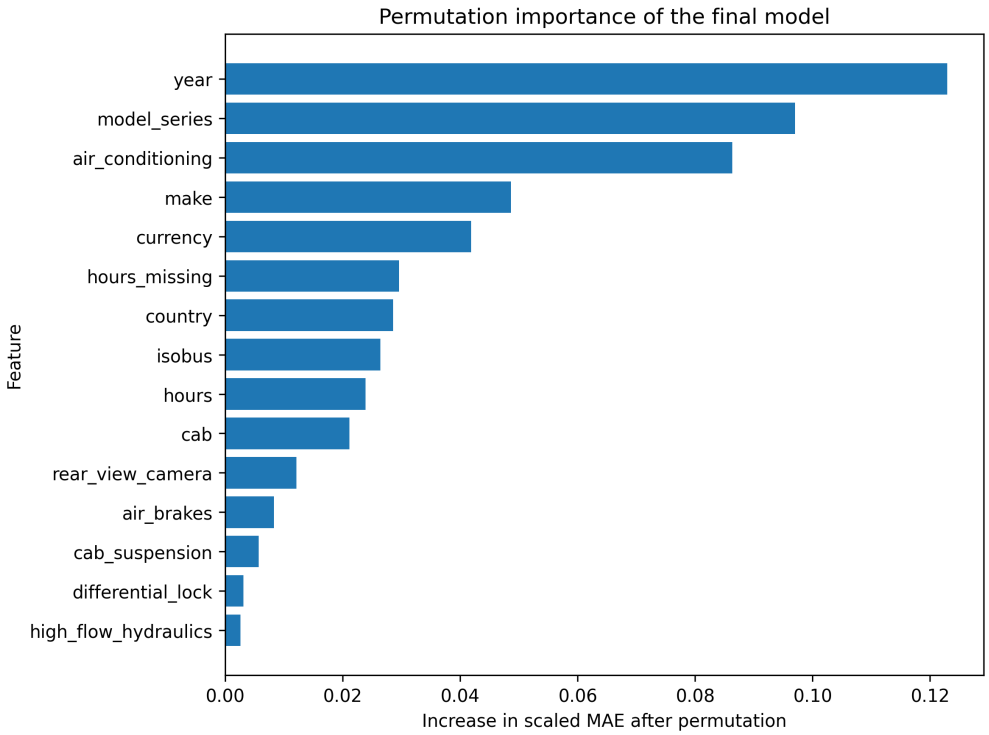


Figure 15: Permutation importance for the selected CatBoost model. Larger values mean that shuffling the feature caused a larger deterioration in predictive accuracy. Importance is predictive, not causal.

The prominence of air conditioning should therefore not be read as an isolated price premium. It may proxy for newer, higher-specification or more thoroughly documented tractors, and correlated predictors can share or redistribute importance. The result is more useful as evidence that structured specification data matters than as a

causal ranking of individual equipment options.

5 Discussion and recommendations

5.1 What the model results mean in practice

CatBoost reduces overall test MedAPE from 40.2% for quadratic regression to 32.0%, corresponding to a 20.5% reduction in typical percentage error. Since the quadratic regression already uses manufacture year, hours, categorical tractor characteristics and equipment indicators, this improvement suggests that the additional nonlinear relationships captured by CatBoost are useful for prediction. CatBoost also performs better in four of the five currency groups, with the small AUD sample as the exception.

Native-currency MAE provides useful monetary context but should be interpreted alongside MedAPE and the local price distribution. This is particularly clear for EUR and GBP: their numerical MAEs are similar, but MedAPE is 18.0% for EUR and 57.5% for GBP because the GBP tractors in the sample have much lower typical prices.

The more important limitation is the information available to the model. Accuracy falls substantially for rare or previously unseen model series, while several potentially useful specification fields are poorly recorded. Further improvements may therefore depend as much on better data coverage as on additional model tuning.

5.2 Horsepower and transmission

Horsepower and transmission both show potentially useful relationships with price, but they are recorded for only 12.9% and 16.0% of tractors, respectively. Using either variable in the current model would therefore require extensive assumptions about the missing values. Transmission also overlaps strongly with manufacturer, model series and overall tractor specification, making its individual contribution difficult to separate.

The practical recommendation is to improve the collection of these fields rather than impute most of their values. Where possible, horsepower and transmission could be recorded systematically at ingestion or obtained from a reliable model-specification reference source. Their additional predictive value could then be assessed using the same validation procedure used for the current models.

5.3 Comparison with existing research

Previous tractor-pricing studies also identify machine characteristics and usage as important predictors of price (Diekmann et al., 2008; Feuz, 2023), while Pearson et al. (2026) reports nonlinear relationships between usage and auction price. These findings are broadly consistent with the patterns observed here, particularly for manufacture year, usage, model identity and specification.

Differences in data sources, markets and available variables mean that numerical results are not directly comparable across studies. The literature is therefore used mainly to place the findings and modelling choices in a wider context.

5.4 Limitations

Several limitations affect interpretation and practical use:

- **No reliable sale date.** The available `observed_date` is a scrape date, preventing reliable modelling of market trends, seasonality or time-based validation.
- **Limited horsepower and transmission coverage.** These fields appear informative but are missing for 87.1% and 84.0% of observations, respectively.
- **Sparse model series.** More than five thousand model series appear only once, and prediction accuracy is substantially lower for rare or unseen series.

- **Equipment recording is incomplete.** For the Boolean equipment fields, a blank value cannot distinguish between a feature being absent and simply not being reported. These variables should therefore be interpreted as indicators that a feature was recorded.
- **Different market composition across currencies.** Currency groups differ in sample size, tractor mix and price range, so differences in predictive accuracy may partly reflect the underlying data available in each market.
- **The model is fitted to realised auction sales.** Marketplace listings were excluded because their recorded values were predominantly asking prices. The results should therefore not be assumed to apply directly to marketplace asking prices or unobserved final transaction prices.
- **Unobserved tractor condition.** Maintenance history, cosmetic condition, attachments and other sale-specific information are not consistently represented in the structured data.
- **Single fixed validation split.** The holdout design protects the test set, but repeated or time-based validation would provide a fuller assessment of variability if reliable sale dates became available.

5.5 Practical recommendations

1. **Include a model-series reliability indicator.** Predictions for rare or unseen series should be clearly identified because these observations have materially larger errors.
2. **Improve horsepower and transmission coverage.** Both variables show useful price relationships but are currently too incomplete for reliable modelling.
3. **Capture a true sale date.** This would allow market trends, seasonality and time-based model validation to be incorporated.
4. **Report both monetary and relative errors.** Native-currency MAE should be used to describe the monetary size of errors within a market, while MedAPE and percentage bands should be used for comparisons across currencies.
5. **Retain quadratic regression as the main benchmark.** Future models should demonstrate an improvement over this fitted reference model before additional complexity is adopted.
6. **Use predictions as decision support.** The model is most reliable for well-represented tractor models with good data coverage; unusual, rare or high-value cases should continue to receive additional review.

6 Conclusion

This project developed a reproducible workflow for understanding and predicting realised used-tractor auction prices. The analysis combined careful data cleaning and EDA with a structured comparison of regression and tree-based models, while keeping the final evaluation in measures that are interpretable in the original price scale.

On the held-out test sample, CatBoost reduced MedAPE from 40.2% for quadratic regression to 32.0%, an 8.2 percentage-point improvement equivalent to a 20.5% reduction in typical percentage error. CatBoost also performed better in four of the five currency groups. However, prediction accuracy was considerably lower for rare or unseen model series, showing that the amount of historical information available for a tractor remains an important limitation.

The results also highlight areas where better data could improve future performance. Horsepower and transmission both showed useful relationships with price but were too sparsely recorded for inclusion in the current model. Improving these fields, capturing a reliable sale date and increasing historical coverage of less common model series are therefore likely to be more valuable than simply increasing model complexity. The current model is best used as a valuation-support tool, particularly for tractors that are well represented in the historical data.

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Appendices

Word Count

Main-text word count: **4968**

The word count was obtained using Overleaf's TeXcount tool and excludes the title page, table of contents, executive summary, declarations, tables, figures, equations, references and appendices in accordance with the project-report guidelines.

The appendices contain supporting information needed to audit and reproduce the main analysis without interrupting the flow of the report. They document predictor inclusion decisions, additional EDA summaries, validation/tuning detail, complete final test metrics and the reproducibility environment. Results central to the business interpretation remain in the main text; the appendices provide the technical evidence behind them.

A Predictor dictionary and inclusion decisions

Table 7: Main predictors and treatment in the modelling workflow.

Variable/group	Treatment	Reason
Manufacture year	Included numeric	Strong EDA relationship and highest final permutation importance.
Hours	Included numeric	Commercially meaningful; 25.1% missing, handled by training-only median imputation.
Hours missing	Included Boolean	Preserves information that usage was not recorded; missingness itself is predictive.
Manufacturer	Included categorical	Major source of tractor heterogeneity.
Model series	Included categorical	Important model identity information; rare levels grouped using training frequencies.
Country	Included categorical	Captures geographic/data-source differences not fully represented by currency.
Currency	Included categorical and target-scaling group	Price levels differ materially; also required for back-transformation.
Drive	Included categorical	Domain guidance maps missing drive to 4WD.
17 equipment indicators	Included Boolean	Structured listing specifications; blank values treated as false under the project data rule.
Horsepower	EDA only	Strong observed association, but only 12.9% recorded and missingness is non-random.
Transmission	EDA only	Descriptive association, but only 16.0% recorded; categories are sparse and often manufacturer-specific.
Observed date	Excluded	Scrape date rather than reliable auction date.
Scrape identifiers/metadata	Excluded	Describe data collection rather than the tractor sale and would interfere with duplicate removal.

The 17 Boolean equipment indicators included in the model are listed below.

Indicator 1	Indicator 2	Indicator 3
front_pto	front_loader	front_linkage
front_axle_suspension	cab_suspension	cab
air_conditioning	air_brakes	isobus
adblue	creeper	trailer_hitch
load_sensing	differential_lock	dual_tires
rear_view_camera	high_flow_hydraulics	–

B Additional data-quality and EDA summaries

B.1 Specification availability and price composition

Table 8: Specification availability and descriptive median price. Raw median prices combine currencies and are shown only to demonstrate that the recorded and missing subsets have different composition.

Field	Status	n	Sample (%)	Median raw price
Hours	Missing	5,728	25.1	9,250
Hours	Recorded	17,129	74.9	34,000
Horsepower	Missing	19,910	87.1	25,000
Horsepower	Recorded	2,947	12.9	33,000
Transmission	Missing	19,191	84.0	22,500
Transmission	Recorded	3,666	16.0	54,000

B.2 Horsepower quantile summary

Table 9: Horsepower EDA by quantile among tractors with horsepower recorded. Median raw price is not currency-converted and should be used only as a descriptive companion to relative log price.

Median horsepower	n	Median relative log price	Median raw price
75	371	-0.455	17,000
100	465	-0.039	24,500
110	404	-0.063	26,000
120	334	0.139	33,000
137	312	0.292	32,000
155	329	0.677	42,000
210	367	0.791	56,000
375	365	1.354	116,000

B.3 Model-series sparsity

Table 10: Model-series frequency diagnostics.

Measure	Value
Unique model series	7,889
Series appearing once	5,081
Series with at least five observations	917
Rows belonging to series with at least five observations	12,883

C Model tuning and validation detail

The hyperparameter search was intentionally small. The purpose was to compare reasonable candidates for each model family without repeatedly adapting a large search space to the same validation sample.

Table 11: Selected validation configuration for each tree family.

Model	Selected configuration
Decision Tree	max depth 16; minimum leaf size 3; minimum split size 8.
XGBoost	650 trees; learning rate 0.03; max depth 6; minimum child weight 1.
CatBoost	700 iterations; learning rate 0.03; depth 7; L2 leaf regularisation 5.0.

C.1 Validation native-currency MAE

Table 12: Validation MAE by model and currency after back-transformation to actual prices. Values are in the native currency of each column; columns should not be added together.

Model	AUD	CAD	EUR	GBP	USD
Currency median reference	50,231	56,602	20,814	11,614	24,594
Additive linear regression	36,259	38,900	11,155	9,060	15,534
Quadratic linear regression	35,724	38,031	10,878	9,137	15,174
Decision Tree	**22,665**	30,637	9,570	8,638	15,209
XGBoost	25,545	32,966	8,388	7,872	14,014
CatBoost	25,389	**29,901**	**7,556**	**7,533**	**13,341**

CatBoost has the lowest validation MAE for CAD, EUR, GBP and USD. The Decision Tree is lower on the small AUD validation sample, while CatBoost is better on the overall unit-free validation metrics used in the main comparison.

D Detailed final test metrics

Table 13: Additional CatBoost test diagnostics and quadratic-regression benchmark metrics by currency. WAPE is calculated separately within each currency.

Currency	CatBoost RMSE	CatBoost MedAE	Price R^2	CatBoost WAPE (%)	Within 20%	Quadratic MAE	Quadratic WAPE (%)	Quadratic within 30%
AUD	54,103	15,624	0.578	40.8	28.8	30,096	38.8	43.9
CAD	57,812	12,721	0.646	40.9	25.5	39,639	55.5	28.3
EUR	12,926	4,238	0.815	21.9	53.6	9,478	27.3	59.1
GBP	13,657	3,455	0.475	54.8	16.5	10,070	74.8	15.7
USD	27,336	5,865	0.631	34.9	40.7	14,995	40.9	49.4

D.1 Bootstrap uncertainty for the main test comparison

To assess the stability of the main CatBoost–quadratic regression comparison, 10,000 paired bootstrap samples were drawn from the 3,429 test observations. The same resampled observations were used for both models in every replicate.

Table 14: Paired-bootstrap uncertainty for the overall test MedAPE comparison. Confidence intervals are percentile-based 95% intervals from 10,000 bootstrap replicates.

Metric	Estimate	2.5th pct.	97.5th pct.
CatBoost MedAPE (%)	32.0	30.6	33.4
Quadratic regression MedAPE (%)	40.2	38.8	42.2
Improvement (percentage points)	8.2	6.6	10.1
Relative reduction (%)	20.5	16.7	24.2

E Permutation-importance detail

Table 15: Largest permutation importances for the final refitted CatBoost model. Values are increases in internal scaled MAE after shuffling the feature.

Feature	Mean importance
Manufacture year	0.123
Model series	0.097
Air conditioning	0.086
Manufacturer	0.049
Currency	0.042
Hours missing	0.030
Country	0.029
ISOBUS	0.026
Hours	0.024
Cab	0.021

F Reproducibility and submitted outputs

The analysis is organised so that the results can be reproduced from the raw dataset rather than relying on manually entered values.

1. Run `01_data_cleaning_and_eda.ipynb`. This notebook reads `market_listings_new.csv`, applies the cleaning and sample-construction steps, writes `cleaned_tractors.csv`, and produces the EDA tables and figures.
2. Run `02_model_training_and_testing.ipynb`. This notebook recreates the train, validation and test split, fits the regression and tree-based models, performs validation-based model selection, refits the selected model, and produces the final test metrics and prediction outputs.

The executed environment used Python 3.13.5, NumPy 2.3.5, pandas 2.2.3, Matplotlib 3.10.8, scikit-learn 1.8.0, statsmodels 0.14.6, XGBoost 3.1.3 and CatBoost 1.2.8. A fixed random seed of 42 is used for the data split and for stochastic model operations where supported.